

Liming Wang

Yao Class, Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University

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Research Interests

Applied cryptography; payment-channel networks and Bitcoin protocol design; security and economics of LLM multi-agent systems.

Education

Tsinghua University, Beijing

Sept 2024 – present (expected Jul 2028)

B.Eng. in Computer Science, **Yao Class** — founded by Turing Award laureate Andrew C.-C. Yao

Manuscripts Under Review (first author)

COLOPHON: Verifiable Agentic Payment to an Unknown Recipient Set

Liming Wang, Xiaoyu Fan, Wei Xu, Junda Liu

Under review at USENIX Security 2027

First author

- First trustless payment protocol for a runtime-emergent, initially unknown recipient set: before the payment becomes irrevocable, the payer recovers a cryptographic certificate of who was paid and how much, without trusting any intermediary.
- Realized on Bitcoin script extended with two near-future opcodes (CSFS, OP_CAT), at collateral independent of delegation depth and fan-out; an equivocating intermediary is provably the sole net loser.
- At equal collateral and latency, delivers a median +7.8 pp more of the delegated work to providers that can actually do it, and leaves +17.6 pp more honest payments unharmed under fund-freezing attacks at Lightning Network scale; on a real LLM agent marketplace, aborts every payment-diverting deal before settlement while accepting every honest substitution.

No API Key, No Problem: Enabling Agent-Centric Payments via Payment Channels

Liming Wang, Xiaoyu Fan, Wei Xu

Under review at NeurIPS 2026

First author

- Cost-aware payment middleware for multi-agent systems: replaces API-key billing with trust-minimized off-chain micropayments via payment channels, with a credential-isolated supervisor holding exclusive payment authority.
- Profiled 6,589 LLM calls across 270 tasks on three MAS frameworks and three backbone models; no quality degradation, 0.6–1.4% runtime overhead, and 35–250× lower settlement cost than on-chain baselines.

Honors & Awards

- **Chinese Mathematical Olympiad First Prize; National Training Team** (ranked 18th) 2023
- Top 100 Youth Commentators, Xinmin Evening News 2023

Skills

Programming: C/C++, Python, Go, TypeScript, Scala, Solidity

Systems & security: Bitcoin script (Taproot/Schnorr), payment channels, protocol security proofs, game-theoretic analysis, smart contracts

Machine learning: LLM fine-tuning (SFT), RAG pipelines, multi-agent orchestration

Languages: English (fluent; TOEFL 105), Mandarin (native)